

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 1

Question:

For the function $f(x, y) = \frac{y}{x^2}$

1. Find the function's domain and range 1

2. Find the boundary of the function's domain and determine if the domain is an open region, a closed region, or neither. 1

3. Investigate whether the limit

$$\lim_{(x,y) \rightarrow (0,0)} y^2 x$$

exists.

1

Please note: There is no partial marking.